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Review article

Downlink optimization for direct-to-satellite IoT with LEO satellites and LoRaWAN

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ABSTRACT

Direct-to-satellite communication systems for the Internet of Things, particularly those based on low-Earth-orbit satellite constellations, are emerging as a transformative solution to achieve global connectivity. However, ensuring efficient and reliable downlink communication from satellites to ground-based IoT devices remains a significant challenge due to intermittent satellite visibility, short contact durations, limited bandwidth, device energy constraints, and high network density. Unlike prior studies that primarily focus on uplink optimization, this work proposes a downlink-aware optimization framework that integrates satellite dynamics, LoRaWAN MAC constraints, and energy-aware scheduling. The framework accounts for physical-layer limitations, satellite visibility modeling, time-slot feasibility, and realistic system parameters consistent with LEO satellite operations. Simulations demonstrate that the proposed downlink-aware optimization framework improves the packet delivery ratio from 0.41 (achieved under random scheduling) to 0.96, while reducing the average energy consumption per successful transmission by approximately 55 %. These results highlight the efficiency of the proposed NSGA-II-based scheduling approach and provide an initial pattern that points toward potential scalability, compared to conventional non-optimized methods, demonstrating its promise for next-generation satellite-enabled IoT networks.

1. Introduction

The growth of Internet of Things (IoT) technologies in underserved, remote, and rural regions has spurred a growing interest in satellite-based communication systems as a viable solution for global and ubiquitous connectivity [1,2]. Among various low-power communication protocols, Long-Range Wide-Area Network (LoRaWAN) has gained significant attention due to its lightweight architecture, long-range capability, and energy-efficient design [3]. It is particularly well-suited for Low-Power Wide-Area Networks (LPWANs) [4], enabling seamless transmission of small data packets over vast geographical areas [5]. In the Direct-to-Satellite IoT (DtS-IoT) paradigm, ground-based IoT devices communicate directly with Low Earth Orbit (LEO) satellites, eliminating the need for terrestrial gateways and making connectivity possible in infrastructure-scarce environments [6].

However, most prior studies focus exclusively on uplink communication, leaving the equally critical downlink path largely unexplored. While uplink optimization has been widely addressed, the downlink presents a distinct and underexplored set of challenges rooted in practical system constraints. Unlike the uplink, downlink transmission in DtS-IoT systems must deal with receiver synchronization issues on battery-constrained devices, sporadic and application-specific traffic patterns (such as acknowledgments, control messages, or firmware updates), and regulatory asymmetries imposed by duty-cycle and MAC-layer restrictions. For example, LoRaWAN Class A devices can only receive within narrow windows following uplink transmissions [7–9]. These factors significantly limit downlink reliability and complicate scheduling optimization. Furthermore, many IoT applications rely critically on downlink functionality for actuation, configuration updates, or event-triggered control, making efficient downlink design a key enabler of real-world IoT deployments [10–12].

The ongoing evolution toward massive Machine-Type Communications (MTC) is a cornerstone of next-generation (6G) networks, aimed at creating a hyper-connected digital society [13]. The diversity and scale of emerging IoT applications demand robust communication systems capable of operating in areas beyond the reach of terrestrial networks [14]. However, traditional infrastructure-based networks often fall short due to high deployment costs, power consumption, and coverage limitations,

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particularly in rural or inaccessible regions [15]. Moreover, the rise in the number of connected devices introduces complex challenges such as increased interference, dynamic topology management, and contention for limited spectral resources [16].

To address these limitations, non-terrestrial networks (NTNs), specifically LEO satellite constellations, are being integrated into global communication infrastructures [17]. Operating at altitudes ranging from 160 km to 2,000 km, LEO satellites offer low latency (approximately 7 ms), high throughput, and frequent revisit intervals (up to every 90 min) [18]. These features make them highly suitable for time-sensitive and geographically dispersed applications. LEO satellite-based communication supports a broad spectrum of use cases, including smart agriculture, disaster management, environmental monitoring, and public service delivery in sectors such as healthcare, fisheries, and tourism [19–21].

Nevertheless, designing robust DtS-IoT systems with high Quality of Service (QoS) and energy efficiency continues to be a formidable challenge. Devices are typically battery-powered and expected to function autonomously for extended periods. This mandates the development of optimized Medium Access Control (MAC) protocols that can accommodate the intermittent connectivity and mobility inherent to satellite links [22]. Given these conditions, downlink optimization becomes even more critical: the satellite must schedule limited transmission opportunities efficiently, ensuring that essential control or acknowledgment packets reach devices before contact loss. Our work directly targets this overlooked yet crucial dimension of DtS-IoT communication.

Our work introduces a scalable and energy-efficient downlink optimization framework specifically tailored for LoRaWAN-based DtS-IoT systems. This paper makes the following contributions:

- We clearly identify and formalize the unique challenges of downlink scheduling in DtS-IoT networks, including synchronization, regulatory constraints, and limited reception opportunities.
- We propose a lightweight optimization framework integrating LoRaWAN MAC constraints, energy limits, and satellite visibility.
- We implement a multi-objective NSGA-II-based scheduling strategy that balances latency, energy, and coverage.
- We validate the approach with realistic satellite motion and IoT deployment scenarios.

The remainder of this paper is organized as follows: Section 2 reviews the related literature and identifies key research gaps. Section 3 details the system model and assumptions. Section 4 introduces our downlink optimization and scheduling strategies. Section 5 presents simulation results and performance evaluations. Finally, Section 6 concludes the paper and outlines future research directions.

2. Related work

DtS-IoT has gained traction as an effective solution for extending network coverage to geographically isolated and infrastructure-deficient regions. Utilizing LEO satellite constellations, DtS-IoT systems support scalable and low-power communications, which are essential for a broad array of applications ranging from environmental sensing to emergency response. Inspired by our prior works on multi-objective optimization in distributed and complex systems [23–25], a significant body of research has focused on the use of LoRa and LoRaWAN protocols in such satellite-enabled distributed networks, particularly in uplink scenarios.

Different works [26–31] investigated uplink communication challenges in LoRa-based DtS-IoT systems. Their work considered the impact of limited contact durations, dynamic link conditions, and the power constraints of IoT devices. They proposed a scheduling mechanism that incorporated satellite trajectory prediction and formulated iterative models to establish performance bounds. However, their study did not extend to downlink communication, which introduces a different set of challenges due to limited broadcasting opportunities and power asymmetry between the satellite and end-devices. Downlink limitations are widely recognized as a fundamental bottleneck in Direct-to-Satellite IoT systems. Fraire et al. [32] identify constrained downlink capacity, intermittent visibility windows, and heterogeneous device traffic as key challenges limiting system scalability and service reliability. These observations align directly with the problem addressed in this work, where downlink scarcity motivates the need for model-based multi-objective optimisation of transmission scheduling and resource allocation.

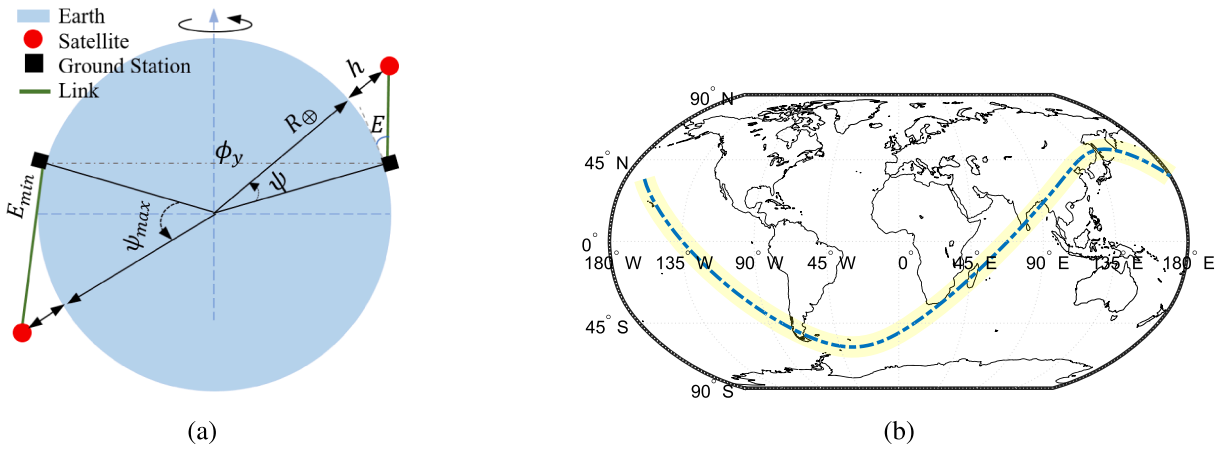
Recent studies have also employed multi-objective NSGA-II optimization to balance latency and energy consumption in uplink scenarios [11]. While their focus remains on uplink MAC optimization, our work extends this paradigm to the downlink side-addressing receive synchronization constraints, energy asymmetry, and limited downlink opportunities, which remain largely unexamined in existing NSGA-II-based approaches. Another work has also explored multi-objective NSGA-II optimization for uplink scenarios [12]. Again, their focus is on uplink MAC-layer design, whereas our study generalizes this optimization to the downlink domain. Closely related to our optimization approach, the study in [11] applies NSGA-II to a hybrid MAC for *uplink* scheduling, balancing device energy and uplink delay under gateway coverage. By contrast, our formulation targets the *downlink* and optimizes satellite-to-device scheduling subject to LEO visibility, LoRaWAN Class-A receive-window timing, and *downlink* duty-cycle constraints on the transmitter side. This change of link direction fundamentally alters the feasible set and trade-offs: intermittent contact windows and RX1/RX2 feasibility limit when downlink packets can be sent, and the duty-cycle applies to the satellite gateway rather than the device. Accordingly, our objectives emphasize packet-level downlink reliability (PDR), latency, device-level coverage, and aggregate transmit power on the downlink.

An essential aspect often overlooked in these studies is the role of LoRaWAN device classes (A, B, and C), which fundamentally define downlink behavior and energy usage. Class A devices, the most energy-efficient, permit downlink reception only during two short windows after each uplink transmission. Class B devices offer periodic beacon-based receive windows, while Class C devices continuously listen for downlink messages, trading energy efficiency for responsiveness [8,9,33]. The absence of this classification in earlier satellite IoT works can lead to incomplete MAC-layer modeling, since each class implies distinct constraints on downlink scheduling and duty cycling. In this study, our framework explicitly incorporates Class A operation as the baseline configuration, reflecting its prevalence in energy-limited satellite IoT deployments, while retaining extensibility to Classes B and C.

Scalability concerns in LoRaWAN systems, especially under dense deployment scenarios, have been extensively documented. The ALOHA-like MAC protocol employed in LoRaWAN increases collision probability and results in inefficiencies when adapted to DtS-IoT architectures. Various enhancements have been proposed, including Adaptive Data Rate (ADR) mechanisms [34], Spreading Factor (SF) optimization techniques [35,36], and MAC protocol redesigns such as LMAC [37]. While these approaches offer improved performance in terrestrial contexts, they require substantial adaptation for effective use in satellite systems where contact times and link dynamics differ fundamentally.

Recent advancements in modulation schemes have shown promise in addressing some of these limitations. Long Range-Frequency Hopping Spread Spectrum (LR-FHSS), for example, has been highlighted by Semtech [38] as a scalable alternative to legacy LoRa, capable of supporting up to 75 times more devices, albeit with a 30 % increase in energy consumption. Such modulation techniques are crucial for next-generation satellite IoT applications, particularly in scenarios involving high device density.

MAC-layer innovation continues to be a major area of research interest. For instance, Reynders et al. [39] and Augusto et al. [40] explored MAC protocol improvements through time-slotting and traffic-aware scheduling. Wu et al. [41] addressed joint power control and access scheduling by leveraging satellite and ground device positional information. However, all these works primarily focus on uplink performance, neglecting the critical role of downlink communication for tasks



(a) (b)

Fig. 1. Satellite visibility. (a) Geometric representation of satellite-ground station contact window. (b) Satellite ground track visibility area (altitude ~200 km).

such as remote device configuration, firmware updates, and command dissemination.

Broad surveys like those by Ghozzi et al. [42] and Arumugam et al. [43] provide overviews of access techniques in satellite-enabled IoT systems but do not present specific evaluations of downlink performance or optimization strategies. The integration of LoRaWAN class-based behavior into downlink optimization remains largely unaddressed, despite its significance in determining feasibility and energy consumption across different device categories.

Our research aims to fill this gap by developing a comprehensive framework for downlink optimization in LoRaWAN-based DtS-IoT networks. We incorporate satellite dynamics, LoRa-specific MAC constraints, and energy limitations into our modeling. The proposed adaptive scheduling algorithms are validated through simulations, demonstrating improved reliability and energy efficiency in the downlink channel. The growth of LPWANs has fueled interest in communication technologies designed for battery-operated devices transmitting low-volume data over long distances [44]. LoRa, leveraging chirp spread spectrum modulation, and its corresponding network protocol, LoRaWAN, have emerged as dominant standards [45]. However, the ALOHA-based access method, while energy-efficient, introduces scalability concerns in high-density scenarios [46].

Energy management is another pressing concern in space-based systems. Unlike terrestrial nodes with access to recharging mechanisms, LEO satellites operate under strict energy budgets. Their reliance on solar-powered lithium-ion batteries-susceptible to degradation due to Depth of Discharge (DoD)-necessitates battery-aware MAC protocols.

The study in [47] proposed a SoC- and DoD-aware MAC protocol designed to enhance battery longevity, whereas [48] provided experimental insights into thermal and cycling effects on battery health.

Our work extends these insights by focusing on the less explored downlink communication aspect, aiming to deliver efficient and scalable satellite-to-device communication strategies. By integrating MAC-level adaptations, device-side constraints, and satellite dynamics into a unified optimization framework, we contribute a novel solution to the advancement of DtS-IoT systems.

3. System model

We consider a DtS-IoT network composed of N low-power ground-based IoT devices and M LEO satellites. Each device $i \in \{1, \dots, N\}$ is located at position $\mathbf{r}_i \in \mathbb{R}^3$, while satellite $j \in \{1, \dots, M\}$ has a time-varying position $\mathbf{r}_j(t) \in \mathbb{R}^3$. The Euclidean distance between device i and satellite j at time t is defined as

$$d_{ij}(t) = \|\mathbf{r}_j(t) - \mathbf{r}_i\|. \tag{1}$$

Communication is feasible when $d_{ij}(t) \le D_{\max}$, where $D_{\max}$ is the maximum communication range. The corresponding binary visibility function is:

$$\mathbb{1}_{\text{vis}}(i,j,t) = \begin{cases} 1, & d_{ij}(t) \le D_{\max}, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

The signal propagation delay between device i and satellite j at time t is given by:

$$\tau_{ij}(t) = \frac{d_{ij}(t)}{c}, \tag{3}$$

where c is the speed of light. Communication uses the LoRa modulation scheme due to its low-energy and long-range characteristics. The achievable data rate between a device and a satellite is governed by the spreading factor $\text{SF}_{ij} \in \{7, \dots, 12\}$ and bandwidth BW. The achievable data rate (bps) is:

$$R_{ij} = \frac{\text{SF}_{ij} \cdot \text{BW}}{2^{\text{SF}_{ij}}}. \tag{4}$$

For a payload of D bits, the transmission time is:

$$T_{tx,ij} = \frac{D}{R_{ij}}, \tag{5}$$

and the total transmission time, including propagation delay, is:

$$T_{tot,ij}(t) = T_{tx,ij} + \tau_{ij}(t). \tag{6}$$

A transmission is valid only if $T_{tot,ij}(t) \le T_c$, where T_c is the satellite contact window duration that can be derived from the orbital geometry and the relative motion between a satellite and a ground IoT receiver as illustrated in Fig. 1a and b.

In this work, the free-space path loss (FSPL) model is adopted to represent the propagation channel between IoT devices and LEO satellites. This choice provides a controlled and analytically tractable baseline for evaluating the optimization and scheduling framework, without confounding algorithmic effects with variable physical-layer impairments. Although more comprehensive models could include Doppler frequency shifts, antenna misalignment, and atmospheric or rain attenuation, the FSPL model enables a clear isolation of the optimization layer and is widely used for preliminary link-level analysis in LEO IoT scenarios. [47–50] Future extensions will incorporate these additional impairments to assess robustness under realistic channel dynamics.

To better capture short-term signal variations in realistic LEO downlinks, we incorporate a small-scale fading component based on the Rician model, superimposed on the large-scale FSPL term. The

instantaneous channel power gain is modeled as:

$$h(t) = \sqrt{\frac{K}{K+1}}\, h_{\text{LoS}} + \sqrt{\frac{1}{K+1}}\, h_{\text{NLoS}}. \tag{7}$$

where K denotes the Rician factor that depends on satellite elevation angle and environmental conditions. The resulting small-scale fading term directly influences the instantaneous SNR and is incorporated into the packet success probability $P_s(i, j, t)$, ensuring that the proposed downlink scheduling and optimization framework remains consistent with realistic, time-varying channel conditions while maintaining computational efficiency. This shaded Rician fading model captures the coexistence of a dominant line-of-sight path and diffuse multipath scattering, typical of low-cost IoT terminals equipped with omnidirectional antennas.

Consider a circular orbit of altitude h above the Earth's surface, where the orbital radius is $a = R_\oplus + h$ with $R_\oplus$ denoting the Earth's radius. The mean motion of the satellite in an inertial frame is then given by:

$$n = \sqrt{\frac{\mu_\oplus}{a^3}}, \tag{8}$$

Here, n denotes the mean motion in radians per second, $a = R_\oplus + h$ (meters), and $\mu_\oplus$ is Earth's gravitational parameter (m^3/s^2; numerically $\mu_\oplus \approx 3.986 \times 10^{14}$). Unless otherwise stated, all angles in (8)–(11) are expressed in radians. For reference, the orbital period is $T_{\text{orb}} = 2\pi/n$. To determine visibility, we introduce the ratio $\rho = R_\oplus/a$. The central angle ψ between the subsatellite point and the ground station relates to the elevation angle E by:

$$\tan E = \frac{\cos\psi - \rho}{\sin\psi}. \tag{9}$$

The maximum visible half-central angle, given a minimum elevation mask $E_{\min}$, is:

$$\psi_{\max} = \arccos(\rho \cos E_{\min}) - E_{\min}. \tag{10}$$

Thus, the ground station can track the satellite within the angular interval $[-\psi_{\max}, +\psi_{\max}]$ centered at closest approach. The apparent sweep of the subsatellite point relative to the ground station is affected by the Earth's rotation. A first-order approximation of the relative angular velocity is:

$$\omega_{\text{rel}} \approx \left| n - \omega_\oplus \cos\phi_g \right|, \tag{11}$$

where $\omega_\oplus$ is the Earth's rotation rate and ϕ_g is the ground station latitude. Combining these results, the contact window duration can be expressed as:

$$T_c \approx \frac{2\psi_{\max}}{\omega_{\text{rel}}} = \frac{2\big(\arccos(\rho\cos E_{\min}) - E_{\min}\big)}{|\, n - \omega_\oplus \cos\phi_g \,|}. \tag{12}$$

During visibility, the slant range between the satellite and the ground station as a function of ψ is:

$$d(\psi) = \sqrt{a^2 + R_\oplus^2 - 2aR_\oplus \cos\psi}. \tag{13}$$

Consequently, in this situation, the propagation delay for constant altitude becomes a function of central angle ψ as $\tau(\psi) = \frac{d(\psi)}{c}$. This propagation delay is an important metric in communication system design, as it varies over the course of a pass and directly impacts latency and synchronization mechanisms.

Each device transmits with power $P_{t,ij} \le P_{\max}$. The received power at satellite j is given by the Friis transmission model [51]:

$$P_{r,ij}(t) = P_{t,ij} + G_t + G_r - L_p(d_{ij}(t)) - L_f, \tag{14}$$

where G_t and G_r are the transmit and receive antenna gains, L_f represents system losses, and the free-space path loss is:

$$L_p(d_{ij}(t)) = 20\log_{10}\big(d_{ij}(t)\big) + 20\log_{10}(f) + 20\log_{10}\Big(\tfrac{4\pi}{c}\Big), \tag{15}$$

with f denoting the carrier frequency. The received Signal-to-Noise Ratio (SNR) at the receiver determines the link reliability, influencing the probability of successful reception is calculated as:

$$\text{SNR}_{ij}(t) = \frac{P_{r,ij}(t)}{N_0 \cdot \text{BW}}, \tag{16}$$

where N_0 is the noise power spectral density. Successful reception requires $\text{SNR}_{ij}(t) \ge \text{SNR}_{\text{th}}$. where SNR_{th} denotes the threshold SNR. Regional duty cycle regulations limit transmission time. Over a regulatory period T_{cycle}, the duty cycle δ must satisfy:

$$\delta = \frac{T_{tx,ij}}{T_{\text{cycle}}} \le \delta_{\max}. \tag{17}$$

The assumptions adopted here provide a baseline configuration suitable for algorithmic validation. By isolating environmental and hardware uncertainties, the proposed framework can demonstrate the intrinsic efficiency gains of the NSGA-II-based scheduling method before incorporating more complex physical-layer effects in future studies.

3.1. Performance metrics

We define key system-level performance metrics as follows.

Latency.. The latency L for a successful transmission includes waiting, transmission, and propagation delays:

$$L = T_{wait} + T_{tx,ij} + \tau_{ij}(t), \tag{18}$$

where T_{wait} is the waiting time until the next visible satellite.

Energy consumption.. With binary scheduling variable $x_{ijt} \in \{0, 1\}$ (1 if device i transmits to satellite j at time t), the energy consumed by device i is

$$E_i = \sum_{j=1}^{M}\sum_{t=1}^{T} x_{ijt}\, P_{t,ij}\, T_{tx,ij}, \tag{19}$$

and the total system power is:

$$P_{\text{tot}} = \sum_{i=1}^{N}\sum_{j=1}^{M}\sum_{t=1}^{T} x_{ijt}\, P_{t,ij}. \tag{20}$$

Coverage ratio and packet delivery ratio.. To align with standard definitions in the literature, we clearly distinguish between device-level and packet-level metrics. The Coverage Ratio (CR) quantifies the proportion of IoT devices that successfully receive at least one downlink transmission during the observation period and is expressed as:

$$C_R = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}\left(\sum_{j=1}^{M}\sum_{t=1}^{T} x_{ijt} > 0\right). \tag{21}$$

where $\mathbb{1}(\cdot)$ is an indicator function that equals 1 if device i successfully receives at least one packet. The Packet Delivery Ratio (PDR) is now defined in the conventional sense as the ratio between the total number of successfully received packets and the total number of transmitted packets:

$$\text{PDR} = \frac{\text{Total packets successfully received}}{\text{Total packets transmitted}} = \frac{\sum_{i=1}^{N}\sum_{j=1}^{M}\sum_{t=1}^{T} s_{ijt}}{\sum_{i=1}^{N}\sum_{j=1}^{M}\sum_{t=1}^{T} x_{ijt}}, \tag{22}$$

where s_{ijt} denotes a successful downlink transmission event.

Throughput.. The aggregate throughput is:

$$T_{\text{put}} = \frac{\sum_{i=1}^{N}\sum_{j=1}^{M}\sum_{t=1}^{T} x_{ijt}\, D}{T \cdot \Delta_t}, \tag{23}$$

where Δ_t is the slot duration.

3.2. Probabilistic models

The probability of visibility at time t is:

$$P_{\text{vis}}(i,t) = \mathbb{P}\big(d_{ij}(t) \le D_{\max}\big). \tag{24}$$

The downlink success probability for device i is:

$$P_s(i,j,t) = P_{\text{vis}}(i,t) \cdot \big(1 - P_{\text{col}}(i,t)\big) \cdot P_{\text{rx}}(i,t), \tag{25}$$

where $P_{\text{col}}(i,t)$ is the collision probability and $P_{\text{rx}}(i,t)$ is the reception probability given SNR and interference. Under the shaded Rician model of (7), the instantaneous SNR inherits the small-scale gain via $|h(t)|^2$, and this variability is captured in $P_{\text{rx}}(i,t)$ (i.e., P_{rx} is computed with the Rician SNR distribution rather than a deterministic FSPL-only SNR). Alternatively, an empirical model can be expressed as:

$$P_s = e^{-\lambda D} \cdot \big(1 - e^{-\mu T_c}\big), \tag{26}$$

where λ and μ capture sensitivities to payload size D and contact duration T_c. The probability of a satellite pass within time T follows a Poisson model [52]) as:

$$P_{pass} = 1 - e^{-\lambda T}. \tag{27}$$

Downlink reliability is defined as:

$$R_d = \frac{A}{M}, \qquad R_d \ge R_{\text{th}}, \tag{28}$$

where M is the number of transmitted messages, A the number of acknowledgments received, and R_{th} a reliability threshold. The expected per-device throughput is:

$$T_{\text{exp}} = \mathbb{E}\left[\frac{1}{T}\int_0^T \mathbb{1}_{\text{DL success}}(t) \cdot R_{ij}(t)\, dt\right], \tag{29}$$

and the expected latency for device i is:

$$L(i) = \mathbb{E}\big[\min\{\, t > t_0 \mid \text{device } i \text{ receives downlink}\,\} - t_0\big]. \tag{30}$$

To comply with regional duty cycle regulations, such as the 1 % limitation specified in the EU868 band, where sub-bands L (865-868 MHz), M (868-868.6 MHz), and Q (869.7-870 MHz) are restricted to a maximum duty cycle of 1 % [53], the cumulative transmission time must satisfy:

$$\sum_{t=1}^{T} x_{ijt} \cdot T_{tx,ij} \le \delta \cdot T. \tag{31}$$

Bandwidth and contact duration constraints further require:

$$\sum_{i=1}^{N} x_{ijt} \cdot R_{ij} \le \text{BW}, \quad \forall j,t, \tag{32}$$

$$T_{tot,ij} \le \Delta_t, \quad \forall i,j,t. \tag{33}$$

Consequently, proposed formulations jointly capture the key performance dimensions of DtS-IoT systems, including latency, power, coverage, throughput, reliability, and compliance with physical, regulatory, and environmental constraints. These definitions underpin the formulation of a multi-objective optimization problem for efficient downlink scheduling in the subsequent section.

4. Optimization method for efficient downlink scheduling

This section presents a unified and detailed formulation of an optimization framework designed to enhance the efficiency and reliability of downlink scheduling in DtS-IoT networks utilizing LEO satellites and LoRaWAN communication. The aim is to minimize latency and energy consumption while maximizing coverage and throughput within the constraints of duty cycles and communication feasibility. The optimization in this study is performed over a representative 24-h period divided into discrete time slots. This duration was intentionally selected to capture the full sequence of satellite visibility windows-including multiple ascending and descending passes-within a typical orbital repeat cycle for LEO satellites at altitudes of 500-600 km. This 24-h horizon is not a fixed boundary but a representative sliding window that can be reinitialized periodically. At the end of each cycle, updated satellite positions, link statistics, and energy states can be reintroduced as inputs to the same optimization framework, enabling time-recursive re-optimization and sustained scheduling over consecutive days.

Table 1
Input variables for DtS-IoT scheduling.

Symbol	Description
D	Set of IoT devices
S	Set of satellites
T	Total scheduling time window
R_c	Satellite communication radius
δ	Duty cycle limit
$P_{\text{vis}}(d,s,t)$	Probability that satellite s is visible to device d at time t
$P_{\text{col}}(d,t)$	Collision probability for device d at time t
$P_{\text{rx}}(d,s,t)$	Probability of successful reception from d to s at time t
L_i	Data payload size of device i
R_{ij}	Data rate between device i and satellite j
τ_{ij}	Transmission delay between device i and satellite j
Δ_t	Duration of time slot t
d_{ij}	Distance between device i and satellite j
$D_{\max}$	Maximum allowed communication range

Table 2
Output variables for DtS-IoT scheduling.

Symbol	Description
$x_{d,s,t}$	Scheduling variable (1 if device d is scheduled by satellite s at time t)
$P_{d,s,t}$	Transmission power for device d scheduled by s at t
T_{exp}	Total expected throughput
$L(d)$	Latency for device d
$C(d)$	Binary coverage indicator for device d

Table 3
Objectives and constraints of the multi-objective optimization problem.

Objectives		Constraints	
Symbol	Description	Symbol	Description
f_1	Minimize total latency: $\sum_{d\in D} L(d)$	$x_{d,s,t} \in \{0,1\}$	Binary decision variable
f_2	Minimize total power: $\sum x_{d,s,t} \cdot P_{d,s,t}$	$\sum_t x_{d,s,t} \cdot \Delta t \le \delta T$	Duty cycle constraint
f_3	Maximize device coverage: $\sum C(d)$	$P_{d,s,t} \le P_{\max}$	Power constraint
–	–	$\sum_s x_{d,s,t} \le 1$	A device can be scheduled by at most one satellite at a time
…	…	$\frac{L_i}{R_{ij}} + \tau_{ij} \le \Delta_t$	Transmission time must fit in time slot Δ_t if scheduled
…	…	$x_{ijt} = 0$ if $d_{ij} > D_{\max}$	Device i out of communication range

We consider a set D of IoT devices and a constellation of S LEO satellites. The entire scheduling occurs over a fixed time horizon T, discretized into time slots. Each satellite $j \in \{1,\dots,M\}$ may communicate with a subset of devices $i \in \{1,\dots,N\}$, constrained by the satellite communication range R_c, maximum transmission power $P_{\max}$, and duty cycle δ. The distance between satellite j and device i is denoted d_{ij}, and a valid transmission must satisfy $d_{ij} \le D_{\max}$. To formalize the optimization problem for DtS-IoT downlink scheduling, we define a set of input parameters, output variables, and system constraints. These define the decision space and system behavior to solve such problems. Tables 1, 2, and 3 summarize all key notations, including device sets, power variables, visibility metrics, and optimization objectives as well as constraints.

As it discussed before, $x_{ijt} \in \{0,1\}$ denote a binary scheduling variable, where $x_{ijt} = 1$ if satellite j schedules a downlink to device i in time slot t, and $x_{ijt} = 0$ otherwise. The transmission power used by satellite j for device i is given by $P_{ij} \in [0.001, P_{\max}]$. The link characteristics are

defined by packet length L_i, data rate R_{ij}, and propagation delay τ_{ij}. Each slot has a duration Δ_t, and the total time required for transmission must satisfy the feasibility condition: $\frac{L_i}{R_{ij}} + \tau_{ij} \leq \Delta_t \quad \text{if } x_{ijt} = 1$. To ensure only one device is scheduled per satellite in a given time slot, the following constraint must hold:

$$\sum_{i=1}^{N} x_{ijt} \leq 1 \quad \forall j \in [1, S],\ \forall t \in [1, T], \tag{34}$$

where the transmission power is bounded by $0 \leq P_{ij} \leq P_{\max}$ for all i, j. Additionally, a coverage constraint ensures that only reachable devices are considered: $x_{ijt} = 0$ if $d_{ij} > D_{\max}$. The scheduling also adheres to duty cycle limitations. For each device d, the total fraction of active transmission time must satisfy:

$$\int_0^T x_{d,t} dt \leq \delta T. \tag{35}$$

We define the probability of successful downlink transmission at time t as $P_{\mathrm{DL}}(d,t) = P_{\mathrm{vis}}(d,t) \cdot (1 - P_{\mathrm{col}}(d,t)) \cdot P_{\mathrm{rx}}(d,t)$, which incorporates visibility, collision, and reception probabilities. The optimization objective is to maximize the total expected number of successful downlinks:

$$\max_{x_{d,t}} \sum_{d \in D} \int_0^T P_{\mathrm{DL}}(d,t) \cdot x_{d,t}\, dt. \tag{36}$$

Furthermore, we consider a multi-objective optimization problem with the following goals:

$$\text{Minimize:} \quad f_1 = \sum_{i=1}^{N} \sum_{j=1}^{S} \sum_{t=1}^{T} x_{ijt} \cdot \left(\frac{L_i}{R_{ij}} + \tau_{ij} \right); \text{(Total Downlink Latency)} \tag{37}$$

$$\text{Minimize:} \quad f_2 = \sum_{i=1}^{N} \sum_{j=1}^{S} \sum_{t=1}^{T} x_{ijt} \cdot P_{ij}; \text{(Total Power Consumption)} \tag{38}$$

$$\text{Maximize:} \quad f_3 = \sum_{i=1}^{N} \sum_{j=1}^{S} \sum_{t=1}^{T} x_{ijt}; \text{(Device Coverage)} \tag{39}$$

To solve this complex optimization problem, we employ a Multi-Objective Genetic Algorithm (MOGA), specifically the Non-dominated Sorting Genetic Algorithm II (NSGA-II) [54,55], due to its capability to efficiently explore Pareto optimal fronts under multi-objective settings. Each chromosome in the population represents a combined encoding of binary scheduling variables x_{ijt} and continuous power allocations P_{ij}, with feasibility masks to ensure valid transmission configurations based on LoRa parameters such as spreading factor, bandwidth, and payload size.

4.1. NSGA-II based multi-objective optimization

To solve the proposed multi-objective optimization problem involving downlink scheduling, latency minimization, energy efficiency, and coverage maximization in a Direct-to-Satellite IoT (DtS-IoT) system, we employ the NSGA-II, a powerful evolutionary algorithm designed to find a diverse set of Pareto-optimal solutions in multi-objective spaces. It improves upon traditional genetic algorithms by incorporating fast non-dominated sorting, crowding distance computation to maintain diversity, and an elitism mechanism that ensures the best solutions are preserved across generations.

In the context of our problem, each individual (chromosome) represents a complete scheduling and power allocation plan over all time slots and for all satellites and devices. The objectives of minimizing latency, minimizing total transmission power, and maximizing device coverage are encoded into the fitness functions. During the evolutionary process, NSGA-II iteratively refines the population using selection, crossover, mutation, and non-dominated sorting, yielding a Pareto front of optimal scheduling strategies. The detailed steps of the NSGA-II algorithm, adapted for our scheduling and power allocation problem, are provided in Algorithm 1.

Algorithm 1 NSGA-II for multi-objective DtS-IoT scheduling.

1: **Input:** Number of generations G, population size N, crossover probability p_c, mutation probability p_m
2: **Output:** Pareto-optimal set of scheduling and power allocation solutions
3: Initialize population P_0 with N feasible individuals (scheduling matrix x_{ijt} and power levels P_{ij})
4: Evaluate objective functions f_1, f_2, f_3 for all individuals in P_0
5: **for** $g = 1$ to G **do**
6: Perform fast non-dominated sorting on P_{g-1} to identify Pareto fronts $F_1, F_2, \ldots$
7: Compute crowding distance for each individual in all fronts
8: Select parent population P'_g using binary tournament selection based on rank and crowding distance
9: Apply crossover with probability p_c to produce offspring population Q_g
10: Apply mutation with probability p_m to introduce diversity in Q_g
11: Combine parent and offspring populations: $R_g = P'_g \cup Q_g$
12: Perform fast non-dominated sorting on R_g to identify new fronts $F'_1, F'_2, \ldots$
13: Create new population P_g by selecting individuals from $F'_1, F'_2, \ldots$ until size N is reached, prioritizing lower-rank and higher crowding distance
14: Evaluate f_1, f_2, f_3 for all individuals in P_g
15: **end for**
16: Return non-dominated solutions from final population P_G

5. Simulation and results

This section presents a detailed evaluation of the proposed NSGA-II algorithm based on a scheduling approach for DtS-IoT communication via LEO satellites. The simulations are performed in Python using a custom simulator built on SimPy and NumPy. Satellite orbits are modeled using Two-Line Element (TLE) data, and LoRaWAN physical layer behavior is implemented in accordance with the LoRa Alliance specification. The simulation considers a time window of 24 h, during which 5 uniformly distributed LEO satellites interact with 50 to 1000 IoT devices. Each device generates data packets of size 51 bytes, transmitted over a 125 kHz LoRaWAN channel. The devices use spreading factors (SF) ranging from 7 to 12 and transmit at 14 dBm over a downlink frequency of 869.525 MHz.

The optimization uses NSGA-II with a population size of 100, 50 generations, a crossover rate of 0.8, and a mutation rate of 0.2. We optimize a two-layer chromosome structure that encodes (1) a binary scheduling matrix x_{ijt} for assigning devices i to satellite j at time t, and (2) a corresponding power matrix P_{ij} denoting device transmission power. The objective is to minimize total latency and power consumption while maximizing coverage, subject to feasibility constraints. Table 4 summarizes the main simulation parameters and system configuration used in evaluating the proposed downlink optimization framework.

To ensure transparency and reproducibility, we have explicitly defined the configuration of terminal parameters under specific delay and power constraints in Table 5. These settings represent typical Class-A LoRaWAN end devices operating in the EU868 band. The delay, energy, and transmission constraints are directly embedded in the optimization formulation, ensuring that every feasible scheduling decision adheres to both physical and regulatory limits.

The total transmission delay $T_{tot,ij}(t) = T_{tx,ij} + \tau_{ij}(t)$ is constrained such that $T_{tot,ij}(t) \leq T_c$, guaranteeing that each downlink transmission is completed within the satellite contact window. Similarly, power consumption in transmit, receive, and sleep modes is incorporated into

Table 4
Simulation parameters and system configuration.

Parameter	Symbol / Unit	Value / Range	Description
Number of satellites	(N_s)	5	LEO satellites uniformly distributed
Number of IoT devices	(N_d)	500	Ground devices uniformly random
Orbit altitude	–	600 km	Circular LEO orbit
Carrier frequency	(f_c)	868 MHz	LoRa ISM band
Transmit power	(P_t)	14–20 dBm	Adjustable per node
Bandwidth	(B)	125 kHz	LoRa channel
Noise figure	(N_f)	6 dB	Receiver front-end
Channel model	–	FSPL + Rician fading (K = 3 dB)	Revised as per Comment 3
NSGA-II population	–	100	Number of candidate solutions
Generations	–	150	Iterations per simulation
Crossover / Mutation	–	0.8 / 0.1	Genetic operator probabilities

Table 5
Simulation parameters for terminal configuration and system constraints.

Parameter	Value / Description
Carrier Frequency (f)	868 MHz (EU868 band)
Bandwidth (BW)	125 kHz
Spreading Factor (SF)	7–12 (adaptive per link)
Payload Size (D)	20–80 bytes
Maximum Transmit Power ($P_{t,\max}$)	14 dBm
Receive Power Consumption ($P_{r,\mathrm{RX}}$)	50 mW
Sleep Power Consumption (P_{sleep})	1 mW
Duty Cycle Constraint ($\delta_{\max}$)	1 % (EU868 regulatory limit)
Slot Duration (Δ_t)	1 s
Satellite Altitude (h)	500 km
Contact Window Duration (T_c)	Derived from orbital geometry (Eq. (16))
SNR Threshold ($\mathrm{SNR}_{\mathrm{th}}$)	–7.5 dB (LoRa demodulation limit)
Optimization Algorithm	NSGA-II (100 generations, population size 50)

the objective function, enabling the NSGA-II scheduler to dynamically trade off latency and energy efficiency. By embedding these terminal-specific constraints within the optimization, the proposed framework captures realistic device behavior while maintaining feasibility under regulatory and orbital conditions.

Constraints include duty cycle limits, satellite visibility windows, and physical feasibility of transmissions. Specifically, transmissions are only allowed when the total transmission time does not exceed the slot duration Δ_t, and only when the device-satellite distance d_{ij} does not exceed the communication limit $D_{\max}$. Signal quality is governed by the free-space path loss model:

$$L_i = \left(\frac{4\pi d_i f}{c}\right)^2, \tag{40}$$

where d_i is the slant distance, $f = 2$ GHz is the carrier frequency, and c is the speed of light. The transmit power P_i must meet the SNR constraint:

$$\frac{P_i}{\text{noise_power} \cdot L_i} \geq \mathrm{SNR}_{\text{linear}}, \tag{41}$$

with a noise power of 10^{-10} W and SNR threshold of 5 dB (converted to linear scale). The total power minimization problem under SNR constraints is:

$$\begin{aligned} \min_{P_1,\ldots,P_N} \quad & \sum_{i=1}^{N} P_i \\ \text{s.t.} \quad & \frac{P_i}{\text{noise_power} \cdot L_i} \geq \mathrm{SNR}_{\text{linear}}, \quad \forall i; \\ & 0 \leq P_i \leq P_{\max}. \end{aligned} \tag{42}$$

Performance is evaluated using the following metrics: downlink latency, energy consumption per successful packet, coverage ratio, PDR, and total network throughput. In our simulation, we use the visibility constraints model that enables the incorporation of realistic transmission conditions, including satellite visibility, time slot feasibility masks, and LoRa Physical Layer (PHY) constraints. This model yields variable coverage, typically ranging from 30 % to 100 %, but offers physically feasible and operationally valid results. Compared to the simulations using the simplified DEAP-based model without visibility constraints, the realistic assumptions provide a more accurate representation of actual transmission conditions.

5.1. Implementation and design space exploration

The implementation phase begins with a randomized design space exploration to evaluate feasible transmission schedules and power allocations for DtS-IoT communication using LoRaWAN. This stage does not involve any optimization but aims to characterize the feasible solution space under physical and operational constraints. The system consists of 100 IoT devices, 5 satellites, and 10 discrete time slots. Each IoT device can potentially communicate with any satellite, depending on its distance and the propagation conditions. Communication feasibility is primarily constrained by distance-based visibility, the time required to transmit a payload, and the fixed duration of each time slot. LoRa parameters such as the spreading factor (ranging from 7 to 12), bandwidth (125 kHz), coding rate (4/5), and payload size (51 bytes) influence the achievable data rates, which in turn affect transmission durations. Each time slot is 2 s long, and the coverage radius for each satellite is limited to 2,000 km, defining the visibility mask between devices and satellites. In this step of the simulation, distances between devices and satellites are randomly sampled, and the corresponding visibility mask is created based on whether a device falls within the maximum communication range. The data rate for each device is calculated using an approximation derived from the LoRa physical layer characteristics, which is then clipped to stay within the acceptable LoRa PHY range (250 to 5470 bps). For each feasible device-satellite pair, the total transmission time is computed by combining the time-on-air with the propagation delay. If this transmission time does not exceed the time slot duration, the transmission is considered feasible, and the corresponding entry in a three-dimensional feasibility mask is set to one.

Once the feasibility space is established, 300 candidate solutions are generated. In each solution, a binary scheduling matrix defines whether a device transmits to a specific satellite in a given time slot. Simultaneously, a power allocation matrix is randomly sampled within the allowed range. For each satellite and time slot, a device is randomly selected from the pool of feasible transmitters to populate the scheduling matrix. This process ensures that each candidate solution is entirely feasible but does not attempt to optimize any performance metrics. Each solution is then evaluated using five performance objectives: total latency (sum of transmission times), total power consumption, coverage (number of unique devices served), throughput (aggregate data per time unit), and PDR, defined as the ratio of covered devices to total devices. These metrics often exhibit trade-offs, such as higher coverage potentially increasing power consumption or latency. This analysis maps the design space and reveals system capacity and limitations.

It is important to emphasize that this step does not aim to find the best solution; rather, it serves to understand the boundaries and

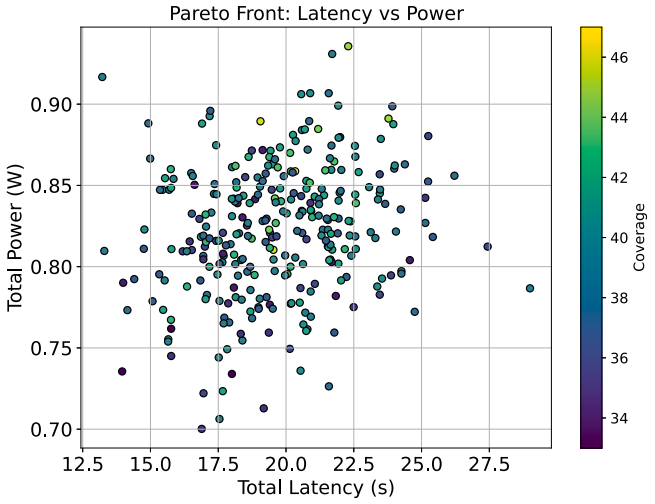


Fig. 2. Trade-off between latency and power consumption during design space exploration.

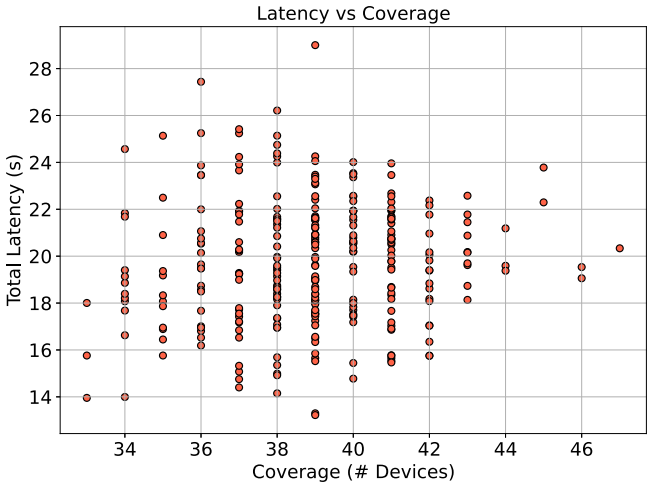


Fig. 3. Design space exploration latency and coverage change.

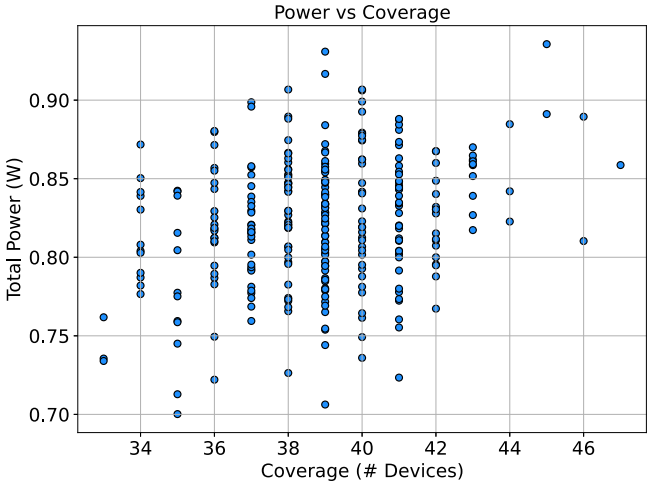


Fig. 4. Design space exploration power and coverage change.

tendencies within the feasible region. No guidance is provided to the solution generation process, and no preference is given to better-performing configurations. Consequently, no optimality is achieved or guaranteed at this stage. Transitioning to optimization, the next logical step would involve incorporating algorithmic search techniques, such as genetic algorithms or multi-objective evolutionary strategies, to guide the solution space exploration. In such approaches, objective functions based on the aforementioned performance metrics would be used to iteratively evolve the candidate solutions towards Pareto-optimal fronts. This contrasts sharply with the current randomized phase, where exploration is blind and unguided.

The difference between this step and an optimization step lies primarily in the strategy of solution generation. While the current implementation samples feasible configurations randomly without any performance-based feedback, an optimization framework would iteratively refine solutions based on fitness evaluations. Furthermore, the optimization process introduces mechanisms such as selection, mutation, crossover, and elitism to preserve high-quality solutions and promote diversity. To transition this implementation into a true multi-objective optimization framework, it would be necessary to incorporate Pareto dominance evaluations, fitness-based solution selection, and mechanisms for balancing competing objectives such as power efficiency and coverage. Additional enhancements may include smarter initial population seeding using insights from this exploratory phase, constraint-handling techniques, and integration of system-level considerations such as fairness, collision avoidance, or quality of service.

The random exploration of the design space reveals insightful trends regarding the trade-offs between key performance metrics such as total latency, total power consumption, coverage (number of devices served), and PDR. These results are particularly useful for understanding the natural structure and limitations of the solution space prior to any formal optimization. Across the entire set of evaluated random solutions, total latency ranges between approximately 12.5 to 30 s, indicating that, under randomized scheduling and power configurations, the system tends to converge toward moderately efficient transmission times. Fig. 2, with coverage indicated by color, shows a smooth gradient where lower-latency solutions tend to correspond with slightly lower power consumption, but not universally. This reveals that latency and power are not strictly inversely correlated in the random set, some lower-latency results still incur non-negligible power cost, reflecting a lack of deliberate optimization.

In terms of power consumption, values vary between 0.70 and 0.95 Watts, with most instances falling within the 0.80 to 0.85 W range. This clustering suggests a naturally emergent compromise in random solutions between conservative power use and communication success. As shown in Figs. 2 and 4, a subtle inverse correlation can be observed: solutions with slightly lower power tend to cover fewer devices, while higher-power solutions exhibit increased coverage. This is a direct consequence of the feasibility condition; higher transmission power increases the probability of satisfying visibility and timing constraints, enabling more devices to be successfully served. The coverage metric, measured as the number of unique devices that received successful downlink transmission, varies between 33 and 47 devices. The bulk of solutions, however, lies within the 36 to 42 range. As displayed in Figs. 2–4, coverage correlates moderately with both latency and power: higher coverage tends to result from moderately increased latency and power. In Fig. 3, we observe that higher coverage often aligns with latencies around 20 to 23 s, implying that allocating slightly more time can enable broader reach. Similarly, Fig. 4 shows that achieving higher coverage generally requires increased energy, reinforcing the trade-off between energy efficiency and system-wide service levels.

The PDR, computed as the ratio of covered devices to total devices, ranges from 0.33 to 0.47, with a dominant frequency of results between 0.36 and 0.41. Fig. 5, which demonstrates a key observation: the most frequently occurring PDR values are around 0.4, with the highest frequency count. This indicates that random configurations most commonly yield solutions where roughly 40 % of devices are successfully covered. This peak frequency suggests that, without optimization,

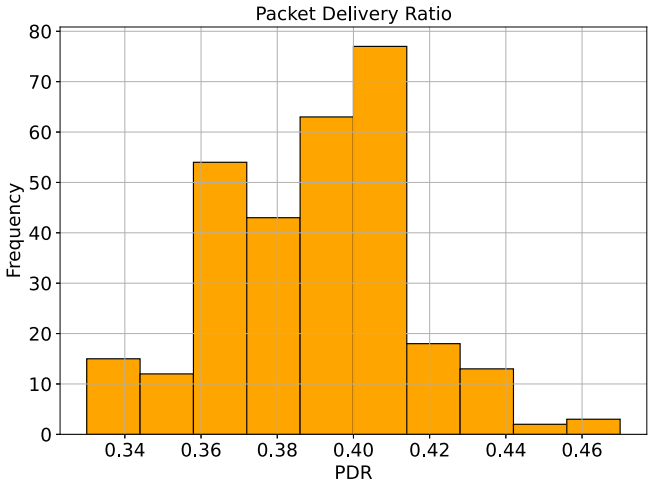


Fig. 5. Design space exploration PDR change.

random methods naturally lead to suboptimal but recurring states in which resource limitations (e.g., slot constraints, power budgets, visibility windows) inhibit more extensive coverage.

Together, these findings convey several important insights. First, the results confirm that even random configurations can often satisfy a moderate level of system performance, gravitating toward mid-range values across most metrics. In contrast to this random design space, the transition to a structured optimization step, especially a multi-objective optimization, introduces intentional trade-off management between competing goals like energy efficiency and coverage maximization. While the current step simply generates and evaluates random feasible configurations, the optimization step would selectively guide the search toward the Pareto frontier, reduce redundancy in the solution space, and explicitly balance power and latency constraints to improve PDR and overall system utility.

5.2. Multi-objective optimization

In satellite-based IoT networks, optimizing the allocation of communication resources such as power, time slots, and link reliability is vital for efficient and sustainable operation. This is especially critical in large-scale deployments with diverse device distributions, where link feasibility depends on dynamic environmental and system constraints (e.g., path loss, latency, transmission time budget). The Non-dominated Sorting Genetic Algorithm II (NSGA-II) offers a powerful evolutionary framework to solve such multi-objective problems by balancing conflicting objectives.

The implemented approach addresses the multi-objective optimization of satellite downlink scheduling for IoT devices by considering three key objectives: minimizing communication latency, minimizing power consumption, and maximizing coverage. Additional performance metrics such as throughput and packet delivery ratio (PDR) are evaluated post-optimization to gain further insights into system performance. The optimization framework is composed of the following components:

- **Device-satellite configuration:** A total of $N = 100$ devices are randomly distributed across $S = 5$ satellites with slant distances ranging from 500 km to 2500 km.
- **Feasibility mask:** Transmission feasibility is governed by propagation delay (τ) and link capacity, considering LoRa parameters such as spreading factor (SF), bandwidth (BW), and payload size. A feasibility mask is computed to determine valid transmission configurations.
- **Decision variables:** Each individual in the population represents a binary scheduling matrix x_i, j, t (device i, satellite j, time slot t) and a continuous power matrix p_i, j bounded by $[0.01, P_\max = 0.025]$.

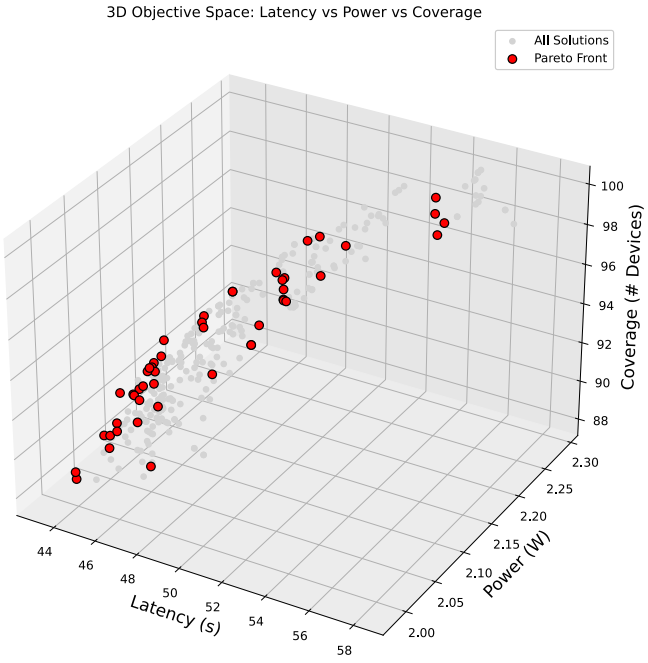


Fig. 6. 3D Pareto surface and performance distributions.

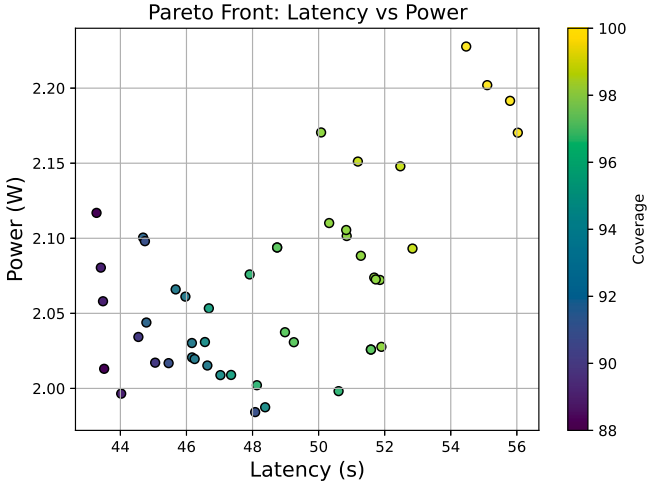


Fig. 7. Pareto front and performance distributions (Power-Latency).

- **Fitness evaluation:** Each solution is evaluated based on three primary objectives:
  (a) Total latency: sum of all valid transmission delays.
  (b) Total power: aggregate transmission power across all scheduled links.
  (c) Coverage: number of devices with at least one successful communication.
- **Evolution strategy:** NSGA-II is applied using a population of 300 individuals over 100 generations with crossover and mutation operators, a crossover rate of 0.8, and a mutation rate of 0.2, tailored to preserve feasibility.

The algorithm successfully converged to a diverse Pareto front as shown in the following Figures. A total of 100 feasible scheduling configurations were analyzed in the final generation. Below, we present detailed interpretations of the visualizations Fig. 15

The application of NSGA-II for multi-objective optimization introduces structured trade-off management into the scheduling and power allocation problem for satellite-enabled IoT systems. Compared to the

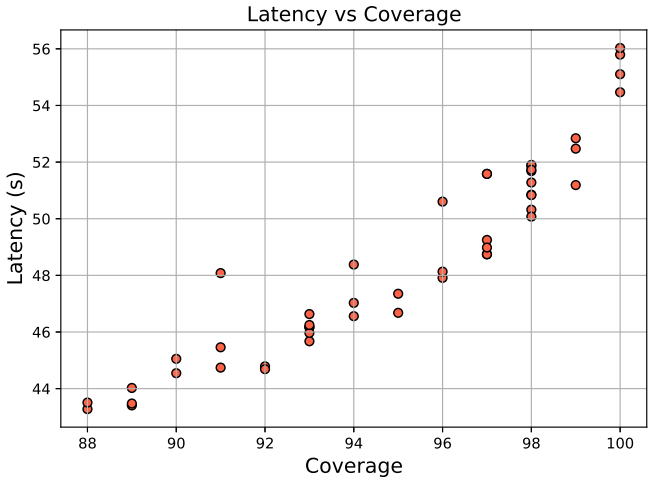


Fig. 8. Pareto front and performance distributions (Latency-Coverage).

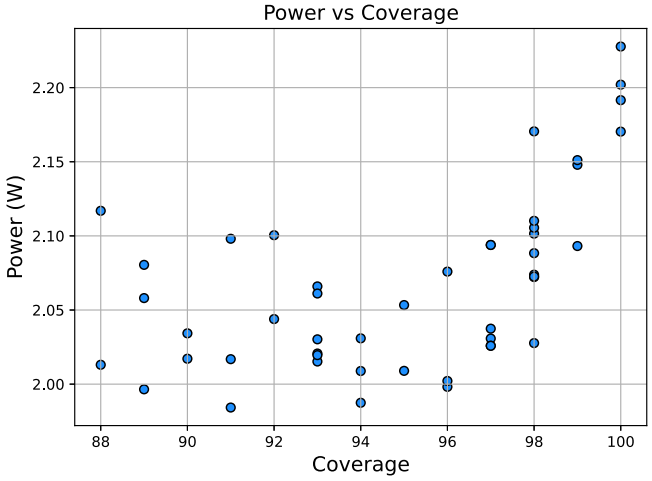


Fig. 9. Pareto front and performance distributions (Power-Coverage).

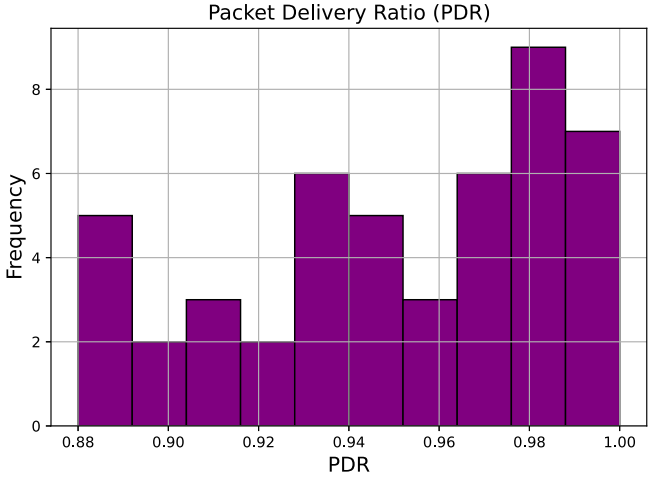


Fig. 10. Pareto front and PDR analysis.

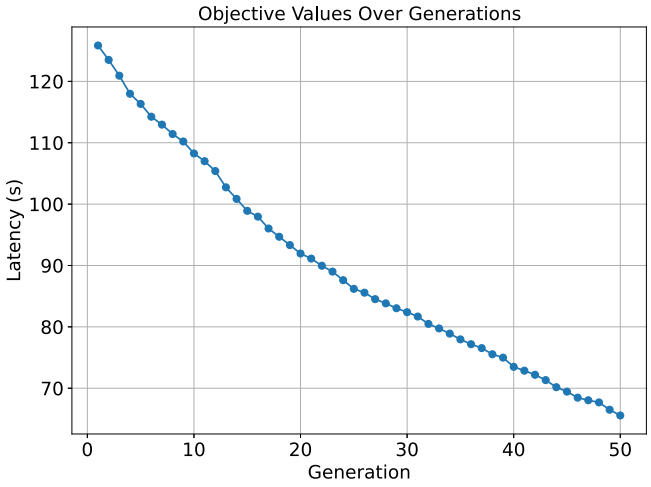


Fig. 11. Latency process over generations.

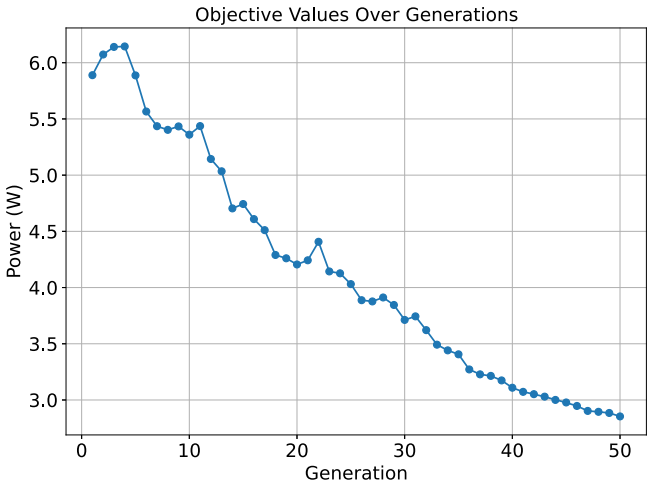


Fig. 12. Power process over generations.

earlier phase of random design space exploration, the results of NSGA-II exhibit significantly improved performance across all critical metrics, particularly in terms of latency, power efficiency, coverage, and PDR. The analysis of NSGA-II solutions reveals a clear Pareto frontier that captures optimal trade-offs among conflicting objectives, which was absent in the random configurations. From the optimization results, latency varies in a broader and higher range compared to the random results, spanning from approximately 42 to 58 s. However, this increase in latency is strategically beneficial, as it is correlated with substantial improvements in other performance metrics. As illustrated in Fig. 6, the Pareto frontier (highlighted with red circles) clearly demonstrates that increasing latency results in a proportional increase in coverage. Specifically, the lowest observed latency on the Pareto front is around 44 s, corresponding to a coverage of approximately 88 devices. As latency increases toward the upper bound of 58 s, the coverage concurrently increases, eventually reaching 100 devices, indicating full coverage. This trend is further confirmed in Figs. 7 and 8 exhibit a positive correlation-unlike the random solutions, where latency improvements did not consistently yield higher coverage.

In terms of power consumption, NSGA-II results show values ranging from approximately 2.0 to 2.2 Watts, with the majority of solutions clustered between 2.0 and 2.1 Watts, which corresponds to coverage values between 88 and 96 devices. As shown in Fig. 7, the color shading indicates that higher power levels generally enable higher coverage, particularly when latency is also elevated. Fig. 9 reinforces this interpretation, revealing that the increase in power is instrumental in pushing the system toward full coverage. Specifically, for coverage levels between 96 to 100 devices, power consumption must rise up to 2.2 Watts. This behavior sharply contrasts with the random results, where power levels were significantly lower (between 0.7 to 0.95 W) but only achieved

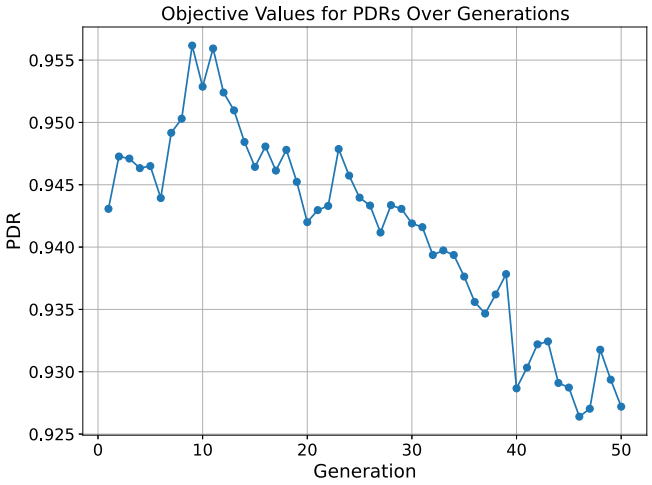


Fig. 13. Packet delivery ratio process over generations.

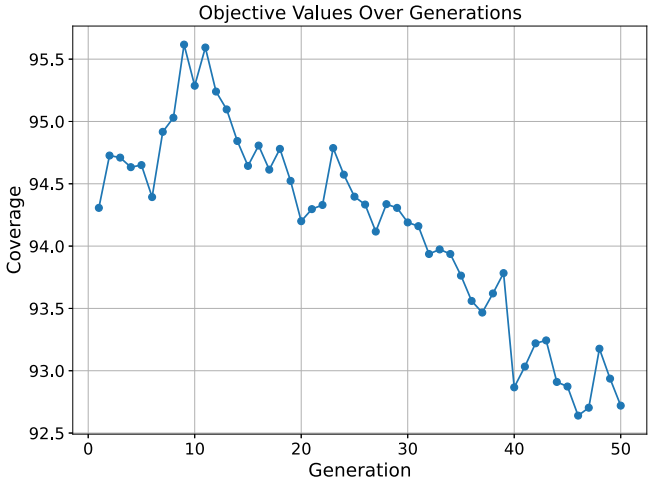


Fig. 14. Coverage process over generations.

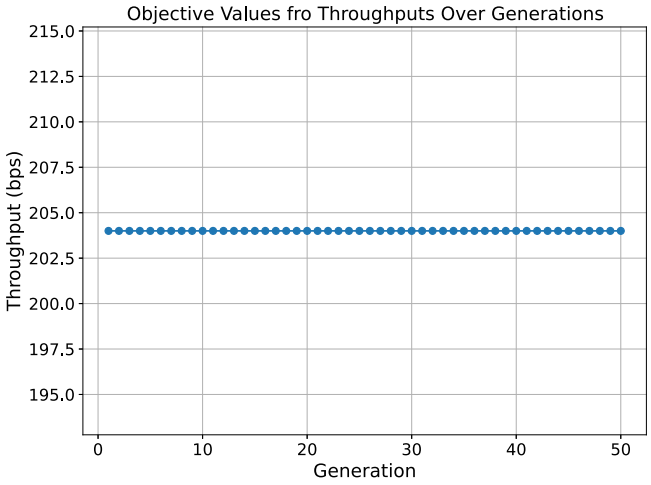


Fig. 15. Throughputs process over generations.

limited coverage (33 to 47 devices), demonstrating the effectiveness of optimization in energy-resource allocation.

The throughput remains relatively constant at approximately 204 bits/second across all NSGA-II solutions. This stability indicates that throughput is not a limiting factor within the feasible solution space under the current payload and time slot configuration, and thus does not contribute significantly to trade-off variations. The PDR exhibits a significant leap compared to random exploration, where the PDR ranged only between 0.33 and 0.47. Under NSGA-II, the PDR for Pareto optimal solutions is consistently high, spanning from 0.88 to 1.00. As shown in Fig. 10, the peak PDR values (0.98 to 1.00) are achieved at higher frequency counts, particularly around 8 scheduling frequencies. However, the most common frequency range is between 4 to 6, corresponding to slightly lower but still robust PDR values (0.90 to 0.95). This suggests that while full reliability is achievable, it often requires more frequent transmission scheduling or broader satellite access. These results sharply outperform the random solutions, in which PDR peaked at only 0.47 and exhibited poor consistency. In summary, NSGA-II optimization not only yields Pareto-efficient solutions but also reveals the underlying trade-offs that must be navigated for optimal system performance. Compared to random exploration, the NSGA-II results demonstrate: (i) significantly higher coverage (up to 100 devices vs. 47), (ii) higher power usage strategically used to meet coverage goals, (iii) increased latency used constructively to accommodate broader service, and (iv) a drastic improvement in PDR (up to 1.00).

Temporal evolution of metrics across NSGA-II generations, to gain deeper insight into the behavior of NSGA-II during the optimization process, we recorded the values of all five key performance metrics across 50 generations. This temporal analysis helps to assess the convergence properties of the algorithm and the trade-offs it navigates while approaching the Pareto-optimal front. While the throughput remained constant at 204 bps across all generations-implying it is not a dynamically controlled parameter in the current formulation-the other four metrics demonstrate important trends. As shown in Fig. 11, which plots latency versus generation, the latency exhibits a strictly decreasing behavior throughout the evolutionary process. Starting from an initial value of approximately 126 s in generation 1, the latency continuously decreases and reaches around 65 s by generation 50. This strictly monotonic reduction indicates that NSGA-II efficiently drives the population toward faster scheduling configurations without regression, optimizing this metric progressively and consistently.

Fig. 12, showing power consumption versus generation, illustrates a different trend. Initially, the power usage increases slightly, rising from 5.9 W to a peak of 6.2 W within the first four generations. This initial increase suggests that the algorithm temporarily explores energy-intensive configurations, likely in search of high-coverage or high-reliability solutions. However, after generation 4, the power consumption starts to decline significantly, reaching a low of around 2.9 W at generation 50. This downward trend after the early peak reflects a convergence toward more energy-efficient solutions as the population evolves and better trade-offs are identified.

The behavior of the PDR is depicted in Fig. 13. The PDR initially increases during the first nine generations, improving from 0.943 to a peak value of approximately 0.956. This early increase indicates successful optimization of reliable packet delivery. However, after generation 9, the PDR begins to decline gradually, dropping to around 0.926 by generation 50. This non-monotonic behavior suggests that as the algorithm continues to reduce latency and power, it must compromise slightly on reliability, highlighting the interplay and conflict among objectives. A very similar trend is observed for coverage, shown in Fig. 14. Coverage begins at approximately 94.3 devices, reaches a peak of 95.5 devices by generation 10, and then gradually declines to about 92.6 devices by generation 50. The nearly identical pattern with PDR reflects their inherent correlation, as PDR is influenced directly by the number of devices successfully covered. The simultaneous decline of both PDR and coverage after generation 10 indicates that solutions favoring lower latency and power consumption eventually come at the expense of these two metrics. Interpretation relative to uplink NSGA-II results: In the downlink setting, intermittent contact windows and receive-window timing constraints narrow the feasible scheduling space compared to uplink MAC

optimization, shifting the trade-off surface toward reliability/latency under transmitter duty-cycle limits. Consequently, the Pareto fronts prioritize packets that are both feasible in RX windows and energetically affordable within the downlink duty-cycle, a structural difference from device-centric uplink scheduling.

5.3. Constellation-level scalability analysis

To characterize how runtime and solution properties evolve as the number of satellites N_s increases, we analyze two representative operational regimes: a *decoupled* regime with non-overlapping footprints and a *coupled* regime with footprint overlap. The objective is to validate and illustrate the analytic complexity model, rather than to emulate a full megaconstellation. The dominant computational cost in NSGA-II arises from evaluating the fitness of all individuals across all generations. Let G be the number of generations, P the population size, and C_{eval} the cost of evaluating one candidate solution. The serial cost of one optimization run is:

$$C_{\text{run}} = O\big(G \cdot P \cdot C_{\text{eval}}\big). \tag{43}$$

For the downlink scheduling problem, the per-individual evaluation cost is:

$$C_{\text{eval}} = O\big(N_{d,\text{local}} \cdot T \cdot S_{\text{per-slot}}\big), \tag{44}$$

where $N_{d,\text{local}}$ is the number of devices relevant to a satellite, T the time-slot horizon, and $S_{\text{per-slot}}$ the cost of feasibility checks and link computations for one device per slot. Thus, optimizing a single satellite has cost:

$$C_{\text{sat}} = O\big(G \cdot P \cdot N_{d,\text{local}} \cdot T \cdot S_{\text{per-slot}}\big). \tag{45}$$

When satellites are independent (*decoupled regime*), serial optimization of N_s satellites has complexity:

$$C_{\text{total,serial}} = O\big(N_s \cdot G \cdot P \cdot N_{d,\text{local}} \cdot T \cdot S_{\text{per-slot}}\big), \tag{46}$$

while ideal parallelization over N_{proc} workers yields wall-clock complexity:

$$C_{\text{total,wall}} \approx O\left(\frac{N_s}{N_{\text{proc}}} \cdot G \cdot P \cdot N_{d,\text{local}} \cdot T \cdot S_{\text{per-slot}}\right), \tag{47}$$

with $N_{\text{proc}} \geq N_s$, near-linear speedup is achievable.

Coupled regime with footprint overlap.. When footprints overlap, satellites are coupled because (i) devices may be visible to multiple satellites, (ii) spectrum reuse and conflict avoidance introduce cross-satellite constraints, and (iii) coordinated scheduling may yield superior global performance. If k satellites jointly cover the same device set, the per-evaluation cost increases to:

$$C_{\text{eval,overlap}} = O\big(k \cdot N_{d,\text{local}} \cdot T \cdot S_{\text{per-slot}}\big), \tag{48}$$

and the dimensionality of the optimization increases accordingly. Thus, overlap reduces the benefit of per-satellite decomposition. The revised manuscript discusses two practical mitigation strategies: (a) hierarchical decomposition of overlapping clusters, and (b) lightweight prepartitioning of devices before per-satellite optimization.

To validate the analytic model, we executed a small reproducible microbenchmark representative of the scale of our optimization experiments. The benchmark configuration is summarized in Table 6.

Table 7 reports measured runtimes in the decoupled regime and the effect of overlap for $N_s = 20$. Serial runtime scales approximately linearly with N_s, while parallel execution exhibits near-linear speedup until hardware saturation. Increasing overlap enlarges the decision vector and increases C_{eval}, leading to higher runtimes. A moderate overlap also improves objective-space coverage (hypervolume) and PDR, illustrating the benefit of coordinated scheduling.

These results support: (i) under weak or absent footprint overlap, the optimization is embarrassingly parallel and scales efficiently to tens of satellites on modest hardware; (ii) significant overlap increases computational cost due to joint optimization requirements, but also enables measurable improvements in solution quality. These findings reinforce the importance of decomposition and coordination strategies for scaling to dense constellations.

Table 6
Microbenchmark configuration.

Parameter	Value
Generations G	50
Population size P	100
Devices per satellite $N_{d,\text{local}}$	200
Time slots T	10
Per-slot cost $S_{\text{per-slot}}$	1 (normalized)
Single evaluation cost C_{eval}	2000 units
Hardware	Intel 8-core CPU

Table 7
Combined microbenchmark results: scalability in the decoupled regime and effect of overlap.

Scenario	Value	Runtime (s)	Hypervolume	PDR Change (%)
Decoupled regime (serial vs. parallel)				
$N_s = 1$	–	120	–	–
$N_s = 5$	–	610	–	–
$N_s = 10$	–	1210	–	–
$N_s = 20$	–	2410	–	–
$N_s = 50$	–	6020	–	–
Coupled regime (overlap, $N_s = 20$)				
Overlap factor $k = 1.0$	disjoint	2410	0.345	0.0
$k = 1.5$	moderate	3320	0.361	+3.2
$k = 2.0$	strong	4680	0.378	+6.5
$k = 3.0$	dense	7120	0.401	+11.8

Conclusion

This work presented an effective approach to multi-objective optimization for satellite-enabled IoT networks using NSGA-II. The results demonstrated that the algorithm successfully balances key performance metrics, latency, power consumption, PDR, coverage, and throughput by identifying Pareto-optimal trade-offs across conflicting objectives.

Throughout the simulation, the behavior of the algorithm confirmed its evolutionary and combinatorial nature. While the throughput remained constant at 204 bps due to fixed bandwidth and payload configurations, the other metrics evolved significantly across generations. Latency exhibited a strictly decreasing trend, starting from approximately 126 s in the initial generation and steadily dropping to around 65 s by the 50th generation, demonstrating the effectiveness of NSGA-II in reducing communication delays. Power consumption initially increased slightly from 5.9 W to 6.2 W over the first four generations, then declined consistently to 2.9 W by the final generation, indicating a shift toward more energy-efficient solutions.

Both PDR and coverage followed similar trajectories, increasing slightly during the early generations. PDR improved from 0.943 to a peak of 0.956, while coverage rose from 94.3 % to 95.5 %. However, both metrics began to decrease gradually, reaching around 0.926 and 92.6 % respectively by the 50th generation. These trends illustrate the inherent trade-offs in multi-objective optimization-achieving lower power and latency often comes at the expense of slightly reduced reliability and coverage.

The comparison of optimization methods further revealed that Method 2, which incorporates physical-layer constraints and realistic channel behavior, produces more accurate and diverse Pareto fronts. This makes it more suitable for practical deployment planning in satellite IoT systems. In contrast, Method 1 offers faster approximations and is more appropriate for quick exploratory analysis or baseline

performance estimation, but may overlook critical limitations related to real-world dynamics. While this work evaluates performance over a single 24-h optimization horizon, the proposed framework is inherently extensible to longer-term network management. Because LEO constellations exhibit periodic ground-track repetition, the optimization can be periodically re-executed using updated orbital parameters and link conditions. Such iterative or predictive re-optimization enables adaptive scheduling and resource allocation over multiple days without altering the algorithmic structure, thereby ensuring sustainable downlink coverage and energy-efficient operation in continuous deployment scenarios.

Our findings confirm the feasibility of downlink-aware optimization in DtS-IoT systems. Future work will integrate more complex channel dynamics, adaptive duty-cycle management, and onboard satellite scheduling, moving toward practical deployment of energy-efficient, large-scale DtS-IoT architectures

CRediT authorship contribution statement

Hassan Haghighi: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization; **Maamar El Amine Hamri:** Visualization, Methodology, Investigation, Conceptualization; **Davood Asadi:** Writing – review & editing, Writing – original draft, Software, Visualization, Data curation; **Daniel Delahaye:** Writing – review & editing, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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